



Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

AS FURTHER MATHEMATICS

Paper 2 Statistics

Friday 19 May 2023

Afternoon

Time allowed: 1 hour 30 minutes

Materials

- You must have the AQA Formulae and statistical tables booklet for A-level Mathematics and A-level Further Mathematics.
- You should have a graphical or scientific calculator that meets the requirements of the specification.
- You must ensure you have the other optional Question Paper/Answer Book for which you are entered (**either** Discrete **or** Mechanics). You will have 1 hour 30 minutes to complete **both** papers.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer each question in the space provided for that question. If you require extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do **not** write outside the box around each page or on blank pages.
- Show all necessary working; otherwise marks for method may be lost.
- Do all rough work in this book. Cross through any work that you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 40.

Advice

- Unless stated otherwise, you may quote formulae, without proof, from the booklet.
- You do not necessarily need to use all the space provided.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



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Answer **all** questions in the spaces provided.

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- 1** The continuous random variable X has variance 9

The discrete random variable Y has standard deviation 2 and is independent of X

Find $\text{Var}(X + Y)$

Circle your answer.

[1 mark]

5

11

13

85

- 2** The random variable T has a discrete uniform distribution and takes the values 1, 2, 3, 4 and 5

Find the variance of T

Circle your answer.

[1 mark]

$\frac{1}{5}$

$\frac{4}{3}$

2

$\frac{13}{6}$

Turn over for the next question

Turn over ►



- 5** Rebekah is investigating the distances, X light years, between the Earth and visible stars in the night sky.

She determines the distance between the Earth and a star for a random sample of 100 visible stars.

The summarised results are as follows:

$$\sum x = 35\,522 \quad \text{and} \quad \sum x^2 = 32\,902\,257$$

- 5 (a)** Calculate a 97% confidence interval for the population mean of X , giving your values to the nearest light year.

[5 marks]



5 (b) Mike claims that the population mean is 267 light years.

Rebekah says that the confidence interval supports Mike's claim.

State, with a reason, whether Rebekah is correct.

[1 mark]

Turn over for the next question

Turn over ►



6 An insurance company models the number of motor claims received in 1 day using a Poisson distribution with mean 65

6 (a) Find the probability that the company receives at most 60 motor claims in 1 day.

Give your answer to three decimal places.

[1 mark]

6 (b) The company receives motor claims using a telephone line which is open 24 hours a day.

Find the probability that the company receives exactly 2 motor claims in 1 hour.

Give your answer to three decimal places.

[2 marks]



- 6 (c)** The company models the number of property claims received in 1 day using a Poisson distribution with mean 23

Assume that the number of property claims received is independent of the number of motor claims received.

- 6 (c) (i)** Find the standard deviation of the variable that represents the total number of motor claims and property claims received in 1 day.

Give your answer to three significant figures.

[2 marks]

- 6 (c) (ii)** Find the probability that the company receives a total of more than 90 motor claims and property claims in 1 day.

Give your answer to three significant figures.

[2 marks]

Turn over for the next question

Turn over ►



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- 7 (b)** By considering observed and expected frequencies, interpret in context the association between the show that a customer attends and whether they enjoyed the show.

[2 marks]

Turn over ►



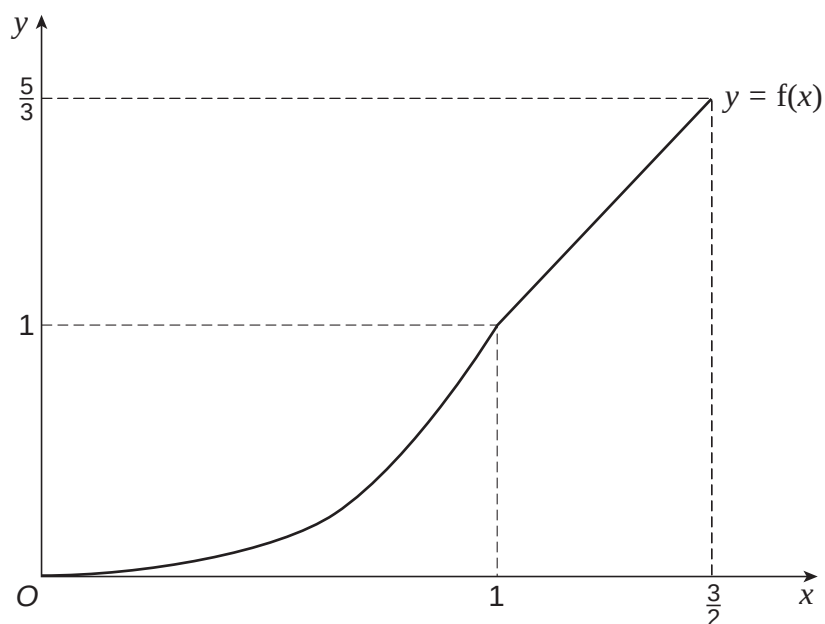
8 The continuous random variable X has probability density function $f(x)$

It is given that $f(x) = x^2$ for $0 \leq x \leq 1$

It is also given that $f(x)$ is a linear function for $1 < x \leq \frac{3}{2}$

For all other values of x , $f(x) = 0$

A sketch of the graph of $y = f(x)$ is shown below.



Show that $\text{Var}(X) = 0.0864$ correct to three significant figures.

[8 marks]



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